

# Types of Hypervisors and Players in Market-19

A hypervisor is the software layer that coordinates virtual machines (VMs). It serves as an interface between the VM and the underlying physical hardware, ensuring that each VM has access to the required physical resources.

The hypervisor also prevents VMs from interfering with each other by isolating memory space and compute cycles.

## Types of Hypervisors

- **Type 1 Hypervisor (Bare-metal):** Runs directly on the physical hardware, replacing the traditional operating system. It is commonly used in virtual server environments where a physical server is partitioned into multiple self-contained segments, each running its own operating system and applications.

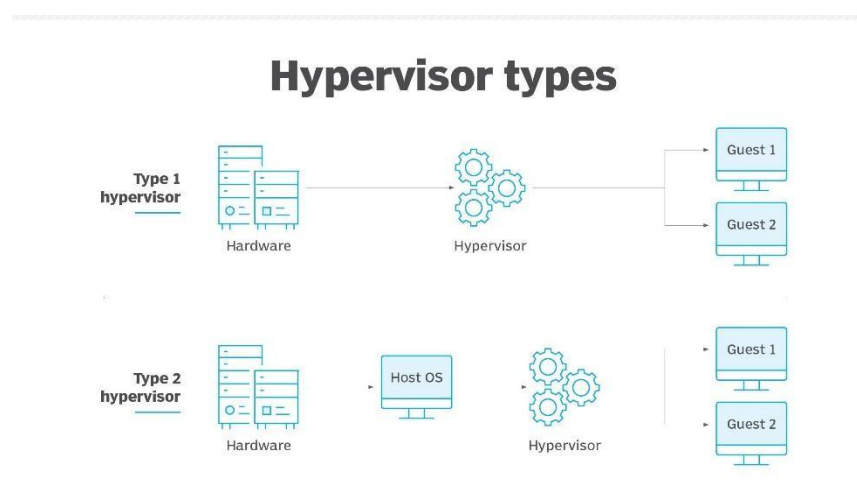
Examples: Microsoft Hyper-V, VMware ESXi, Citrix XenServer.

- **Type 2 Hypervisor (Hosted):** Runs as an application on top of an existing operating system. It is mostly used on endpoint devices to run guest operating systems. Since it depends on the host OS to manage hardware access, it introduces a performance overhead.

Examples: Oracle VirtualBox, VMware Workstation.

- **Hybrid Hypervisor:** Combines features of both Type 1 and Type 2. It allows direct hardware access like a bare-metal hypervisor while also providing flexibility of a hosted hypervisor.

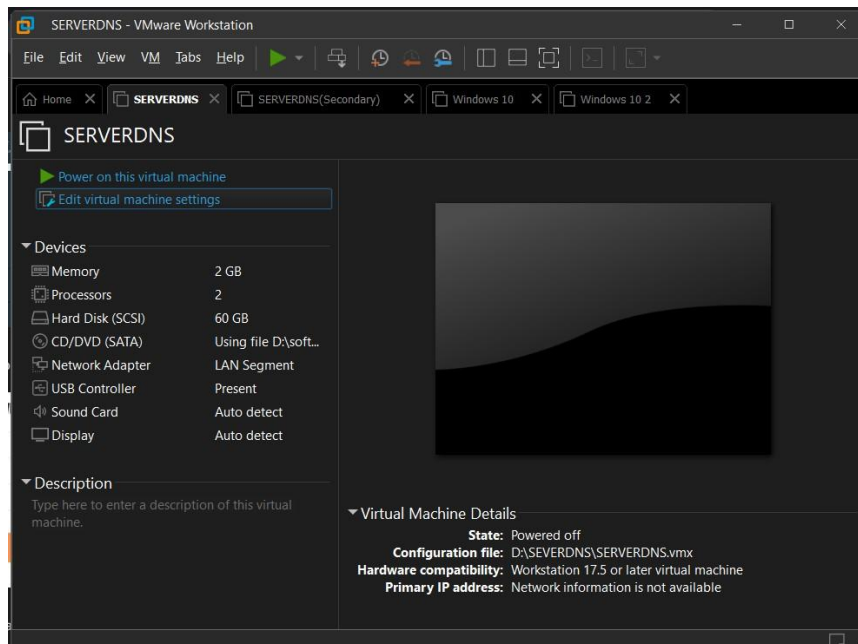
Examples: Windows Subsystem for Linux 2 (WSL2), Apple's Hypervisor framework.



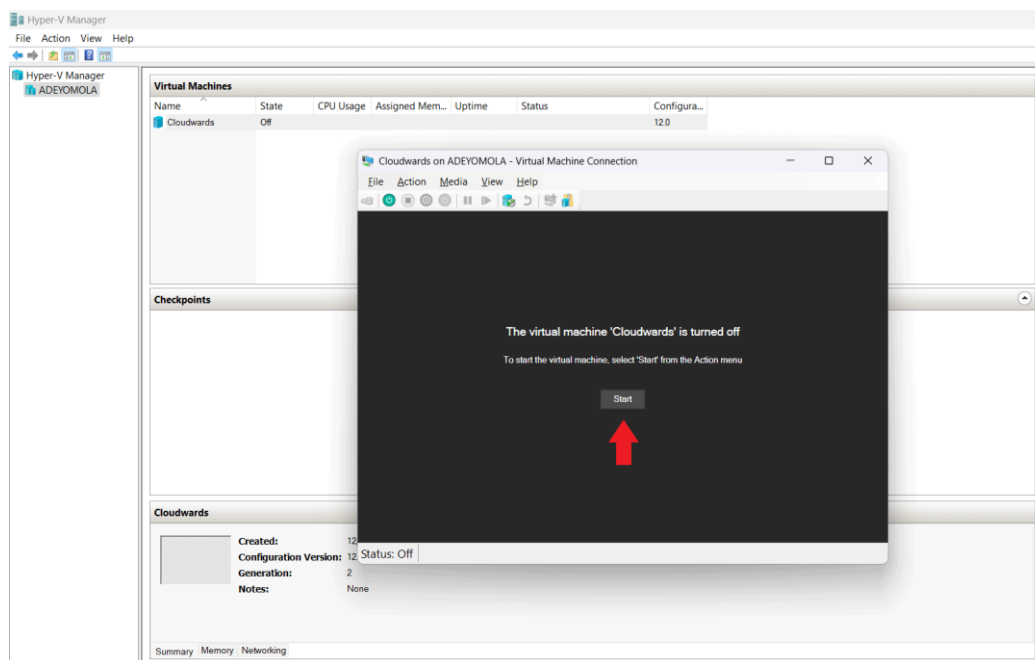
## Examples of Hypervisors

- **VMware – ESXi and vSphere platform**

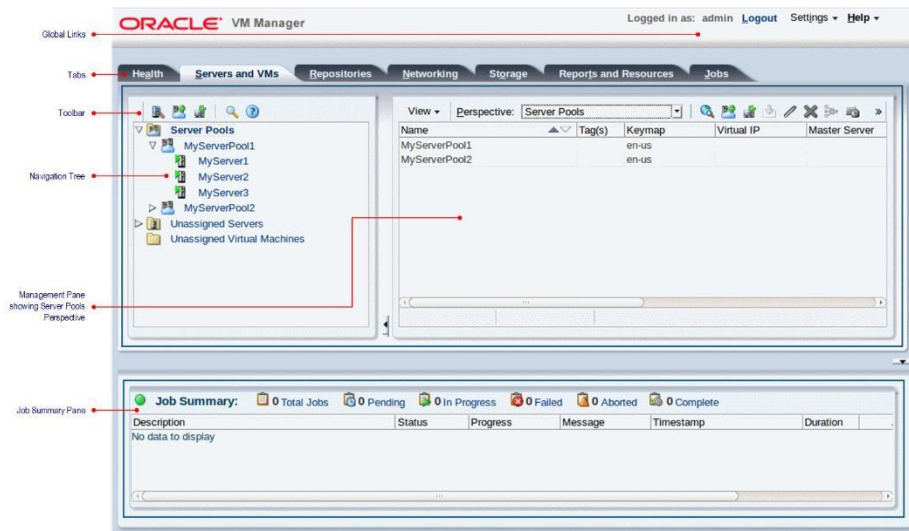
- It is a Type 1 bare-metal hypervisor.
- It installs directly on the hardware without needing an operating system.



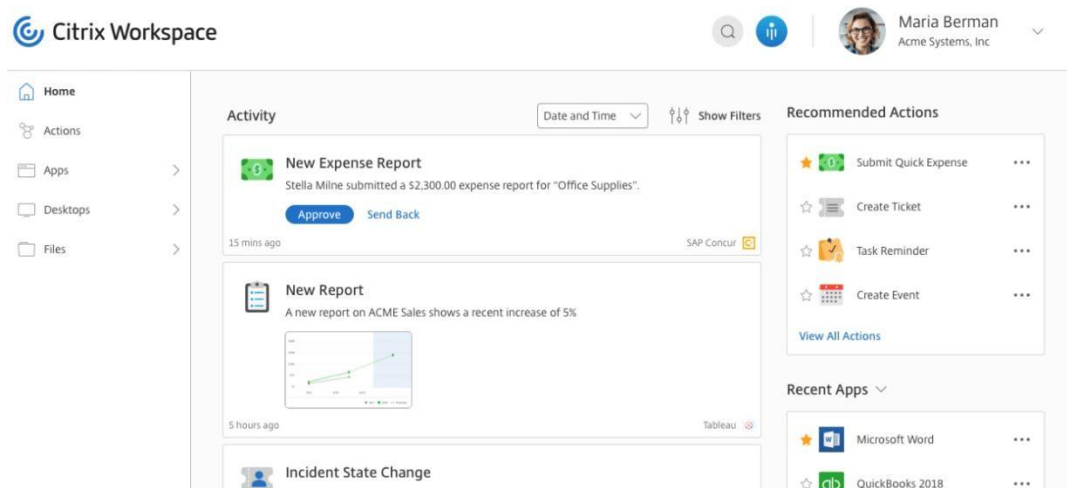
- **Microsoft – Hyper-V.**
- Hyper-V is Microsoft's Type 1 hypervisor built into Windows Server and available on Windows 10/11 Pro and Enterprise editions.
- It allows running multiple Windows and Linux VMs on a single physical host.



- **Oracle – VirtualBox.** • VirtualBox is a Type 2 hosted hypervisor.
- It runs on top of an existing operating system such as Windows, macOS, or Linux.



- **Citrix – XenServer.**
- XenServer is a Type 1 hypervisor based on the open-source Xen Project.
- Provides virtualization for servers and desktops.
- Strong in Virtual Desktop Infrastructure (VDI) solutions.



- **Red Hat – KVM (Kernel-based Virtual Machine).**
- KVM is built into the Linux kernel, turning Linux into a Type 1 hypervisor.
- It supports running both Linux and Windows VMs.

